

# A DANGERous Data Challenge

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DANGER: Data, Geometry, and Numbers, BIRS  
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# AI for math

Artificial intelligence and machine learning tools are used in a variety of different ways to help us gain insight and understanding, generate conjectures, and prove theorems.

- Analyze data
  - Search for patterns and correlations (data science)
  - Classification (model training)
- Construct examples
  - Generate extremal/interesting examples ([Funsearch](#), [PatternBoost](#), [AlphaEvolve](#), ...)
  - Extend known or suspected patterns, test conjectures
- Prove theorems
  - Informally ([First Proof](#) solutions by [OpenAI](#) and [DeepMind/Aletheia](#), [Bryan-Elek-Manners-Salafatinos-Vakil](#), [Swaminathan](#), ...)
  - Formally ([AlphaProof](#), [Aristotle](#), [Ax-Prover](#), [AxiomProver](#), [DeepSeek-Prover](#), [Gauss](#), [Gödel-Prover](#), [Kimina-Prover](#), [Seed-Prover](#), ...)

## It's all 0's and 1's

Mathematics has two key features that make it especially attractive for AI:

- An inexhaustible supply of noiseless data.
- Computational access to ground truth.

What is data?

In mathematics, data are not observations, they are facts (aka theorems). That the elliptic curve  $y^2 + xy + y = x^3 + x^2 - 10x - 10$  has rank 0 is one of 3,824,372 similar facts one can find in the [LMFDB](#). Like all mathematical facts, it can be encoded as a sequence in some fixed alphabet (WLOG, we can assume that alphabet is  $\{0, 1\}$ ).

What is a proof?

A proof (at least in modern mathematics) is a series of permissible steps that allow one to derive a stated conclusion from a stated hypothesis and an agreed list of axioms. Every proof can (at least in principle) be encoded as a sequence in some fixed alphabet.

## Warmup example: compositeness

Fact: 91 is composite.

Proof:  $91 = 7 \times 13$ .

Better proof:  $2^{91} \not\equiv 2 \pmod{91}$

The integer 2 witnesses/certifies that 91 is composite.<sup>1</sup>

The triple  $(91, 7, 13)$  encodes a proof of the compositeness of 91 that can be mechanically translated into a formal proof with a quasi-linear expansion factor.

The pair  $(91, 2)$  also encodes a proof of the compositeness of 91 that can be mechanically translated into a formal proof, with a quasi-quadratic expansion factor.

The singleton  $(91)$  can also be viewed as encoding a proof of the compositeness of 91 that can be mechanically translated into a formal proof, but with an exponential expansion factor:  $2|91 \vee 3|91 \vee 4|91 \vee 5|91 \vee 6|91 \vee 7|91 \vee 8|91 \vee 9|91$ .

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<sup>1</sup>Not all composite numbers have a witness of this type, in general one uses the more general notion of a Rabin-Miller witness.

## Primality proving

Fact: 89 is prime.

Non-proof:  $89 = 1 \times 89$

Non-proof:  $2^{89} \equiv 2 \pmod{89}$

Proof:  $2^{88} \equiv 1 \pmod{89} \wedge 2^{44} \not\equiv 1 \pmod{89} \wedge 2^8 \not\equiv 1 \pmod{89}$

$\wedge 89 - 1 = 88 = 2^3 \times 11 \wedge 2 \text{ is prime} \wedge 11 \text{ is prime}$

Better proof: The point  $P := (6, 45)$  on the elliptic curve  $y^2 = x^3 + 13x^2 + x$  satisfies  $2^4 P \neq 0$  and  $2^5 P = 0$ , with  $2^5 > q + 1 + \lfloor 2\sqrt{q} \rfloor$  where  $q = \lfloor \sqrt{p} \rfloor = 9$ .

The *Pomerance triple*  $(89, 13, 6)$  encodes a proof of the primality of 89 that can be mechanically translated into a formal proof, with a quasi-quadratic expansion factor.

The **Pomerance triple**  $(89, 13, 6)$  encodes a proof of the primality of 89 that can be mechanically translated into a formal proof, with a quasi-quadratic expansion factor. **Pomerance proofs** are the **gold standard** of primality proofs; every other type (Pratt, ECPP, AKS, ...) is more complicated and has a larger expansion factor.

## Pomerance proofs

For a positive integer  $p$ , define  $k_p := \lceil \log_2(q + 1 + 2\sqrt{p}) \rceil$  where  $q = \lfloor \sqrt{p} \rfloor$ .

### Definition

A triple of integers  $(p, A, x_0)$  with  $x_0, A \in [0, p-1]$  and  $A^2 \not\equiv \pm 2 \pmod{p}$  is a **Pomerance triple** if there exist integers  $B, y_0 \in [0, p-1]$  such that  $P := (x_0, y_0)$  is a solution to  $E : By^2 = x^3 + Ax^2 + x$ , with  $2^{k_p}P = 0$  and  $2^{k_p-1}P \neq 0$ .<sup>2</sup>

### Theorem

*If  $(p, A, x_0)$  is a Pomerance triple with  $p > 2$  odd, then  $p$  is prime.*

### Proof.

$A \not\equiv \pm 2 \pmod{p}$  ensures  $E$  is non-singular modulo any odd divisor  $q$  of  $p$ .

If  $p$  is not prime then we may choose  $q \leq \sqrt{p}$ . The point  $P = (x_0, y_0)$  has order  $2^{k_p}$  which is larger than  $\#E(\mathbb{F}_q) \leq q + 1 + 2\sqrt{q}$ , a contradiction. □

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<sup>2</sup>The inequality  $2^{k_p-1}P \neq 0$  must hold over  $\mathbb{Z}/q\mathbb{Z}$  for every  $q|p$ .

## Pomerance proofs

Note that  $By_0^2 = x^3 + Ax^2 + x$  uniquely determines the square class of  $B$  in  $(\mathbb{Z}/p\mathbb{Z})^\times$ , which uniquely determines the isomorphism class of  $By^2 = x^3 + Ax^2 + x$  for a given  $A$ , so we can take  $B$  to be any representative of its square class, and this determines  $y_0$ .

But in fact we never need to know the values of  $B$  and  $y_0$ !

Given a projective point  $P = (x_1 : y_1 : z_1)$  on  $By^2z = x^3 + Ax^2z + xz^2$ , we can compute  $2P = (x_2 : y_2 : z_2)$  by letting  $C := (A + 2)/4$  and applying

$$u = (x_1 + z_1)^2$$

$$v = (x_1 - z_1)^2$$

$$w = u - v$$

$$x_2 = uv$$

$$z_2 = w(v + Cw)$$

## Certifying a Pomerance triple

```
1 def pp_verify(p, A, x0):
2     from math import gcd, isqrt
3
4     if p < 5 or p % 2 == 0:
5         return False
6
7     q = isqrt(p)
8     k = (q + 1 + isqrt(4*q)).bit_length()
9
10    # p not prime ==> prime divisor < q
11    # least k such that 2^k > q + 1 + 2*sqrt(q)
12
13    if gcd(A * A - 4, p) != 1:
14        return False
15    # singular
16
17    X, Z = x0 % p, 1
18    Zprev = None
19
20    # precompute C = (A + 2)/4 mod p for doubling formula
21    C = ((A + 2) * ((p + 1) // 4 if p % 4 == 3 else (3*p + 1) // 4)) % p
22    for i in range(k):
23        Zprev = Z
24        U, V = (X+Z)*(X+Z) % p, (X-Z)*(X-Z) % p
25        W = U - V
26        X, Z = U*V % p, W*(V+C*W) % p
27
28    # if p|Z and (Zprev,p)=1 then (p,A,x0) is valid
29    return Z % p == 0 and gcd(Zprev, p) == 1
```



## Demonstration

Code and data is available in the following GitHub repo:

<https://github.com/AndrewVSutherland/DANGER3>

```
python3 vpp.py 89 13 6
```

```
python3 lean_vpp.py 89 13 6 >pp89.lean
```

Online Lean4/Mathlib calculator available at:

<https://live.lean-lang.org/>

# The landscape of Pomerance triples

## Theorem

*Pomerance triples  $(p, A, x_0)$  exist for every prime  $p > 3$ .*

For any fixed  $p$ , there are on the order of  $\sqrt{p}$  integers  $A \in [0, p-1]$  that appear in a Pomerance proof for  $p$ ,<sup>3</sup> and for each such  $A$  there are  $2^{k_p-1} \approx \sqrt{p}$  possibilities for  $x_0$ . Thus every prime  $p > 3$  appears in roughly  $p$  distinct Pomerance triples.

Given a prefix  $(p, A)$  of a Pomerance triple, it is easy to find an  $x_0$  for which  $(p, A, x_0)$  is a Pomerance triple (this takes quasi-quadratic expected time).

For a general prime  $p$ , no efficient algorithm is known for finding a Pomerance triple  $(p, A, x_0)$ . The fastest known general algorithm simply picks  $A$ 's at random and tries to extend  $(p, A)$  to  $(p, A, x_0)$  (this will fail or succeed in quasi-quadratic expected time).

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<sup>3</sup>"On the order of" ignores factors that are  $O(\log p)$ .

## Data challenge

**Challenge:** Given a (potentially large) prime  $p$ , generate a Pomerance triple  $(p, A, x_0)$ .

Resources available at <https://github.com/AndrewVSutherland/DANGER3>:

- [vpp.py](#): Python program to test candidate triples.
- [pp10.txt](#): the 236,264 Pomerance triples with  $p < 2^{10}$ .
- [pp20.txt](#): one Pomerance triple for the 82,023 primes  $3 < p < 2^{20}$ .
- [pp12.txt.gz](#): the 3,083,880 Pomerance triples with  $p < 2^{12}$ .
- [pp24.txt.gz](#): one Pomerance triple for each of the 1,077,869 primes  $3 < p < 2^{24}$ .
- [pp16A.txt.gz](#): the 4,713,069 prefixes  $(p, A)$  of Pomerance triples with  $p < 2^{16}$ .

Resources available elsewhere (links available in the GitHub repo):

- [pp14.txt.gz](#): the 43,307,624 Pomerance triples with  $p < 2^{12}$  (646Mb).
- [pp16.txt.gz](#): the 598,705,640 Pomerance triples with  $p < 2^{12}$  (9887Mb).
- [pp28.txt.gz](#): one Pomerance triple for 14,630,841 primes  $3 < p < 2^{28}$  (386MB).